



FUNDAMENTALS OF PROBABILITY

QUESTION BANK

MODULE 3

SHORT QUESTION AND ANSWER

1. Define Experiment and Random Experiment

Experiment:

Answer

If an action is repeated under similar and homogeneous conditions which gives predictable and unpredictable results is known as experiment.

Random Experiment:

If the result of experiment is not certain and it is any one of all possible outcomes then that experiment is called Random Experiment.

2. Define probability

Answer:

Probability:-

Let n be mutually exclusive, equally likely and exhaustive events of random experiment, let E be an 'event', m be no of favourable cases of E then

$$P(E) = (\text{no of favourable cases of } E) / (\text{Total no of events})$$

$$\text{Exhaustive events} = n(E) / n(S) = m / n$$

3. Write the statement of Additional theorem

Answer

Let statement :- Let A, B be any 2 events in the sample space S then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

4. In a class there are 10 boys and 5 girls. A Committee of 4 students is to be selected from the class so, find the probability from the committee to contain atleast 3 girls.

Answer

No of boys = 10

number of girls = 5.

The probability that the committee contains atleast 3 girls is:

$$= ({}^5C_3 \times {}^{10}C_1 + {}^5C_4 \times {}^{10}C_0) / {}^{15}C_4$$



Long question and answers

Q1. A class consists of 6 girls and 10 Boys. If a committee of 3 members is chosen at a random from the class. find the probability that

i, 3 boys are selected

ii, Exactly 2 girls are selected.

i) Total no of students = $10 + 6 = 16$.

$$n(S) = 16$$

no of ways of selecting 3 members from 6 girls and 10 Boys is $n(S) = {}^{16}C_3$

Let E be the event of choosing 3 boys $n(E_2) = {}^{10}C_3$

$$\begin{aligned} P(E) &= \frac{n(E_1)}{n(S)} = \frac{{}^{10}C_3}{{}^{16}C_3} \\ &= \frac{\frac{(10 \times 9 \times 8)}{(3 \times 2 \times 1)}}{\frac{(16 \times 15 \times 14)}{(3 \times 2 \times 1)}} = \frac{10 \times 9 \times 8}{16 \times 15 \times 14} = \frac{3}{14} \end{aligned}$$

ii) let E_2 be the event of selecting exactly 2 girls and 1 boys is $n(E_2) = {}^6C_2 \times {}^{10}C_1$

$$P(E_2) = \frac{n(E_2)}{n(S)} = \frac{({}^6C_2 \times {}^{10}C_1)}{{}^{16}C_3} = \frac{15}{26}$$

2Q. Axiomatic definition of probability:-

Let S be a finite sample space where real valued function 'P' defined as $P: S \rightarrow R$ is called probability function.

If all the axioms are satisfied:-

i. Axiom of positivity $P(E) \geq 0$.

ii. Axiom of certainty $P(S) = 1$

iii. Axiom of certainty $P(S) = 1$ iv. Axiom of union:

if E_1, E_2 mutually exclusive events (i.e $E_1 \cap E_2 = \emptyset$)

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

Note:-

If $E_1, E_2, E_3, \dots, E_n$ are mutually,

$$P(E_1 \cup E_2 \cup E_3 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \dots + P(E_n)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



3 Q Two cards are selected at random from 10 and each numbered 1 to 10. find the probability of sum is odd.

- If
- i, Two 2 cards are drawn together**
 - ii. Two cards are drawn one after the other with replacement.**
 - iii. Two cards are drawn without replacement**

Answer

There are 10 cards. Choosing of 2 cards = $n(S) = {}^{10}C_2 = 45$

E be the event of getting sum is odd of two cards:-

5-even 5-odd

$$n(E) = 5C_1 \times 5C_1 = 5 \times 5 = 25$$

$$\text{Probability of getting an odd sum } P(E) = n(E) / n(S) = 25 / 45 = 5 / 9$$

$$(ii) n(S) = {}^{10}C_1 \times {}^{10}C_1 = 10 \times 10 = 100$$

E1 be the event of getting sum is odd

5-even 5-odd

$$n(E1) = 5C_1 \times 5C_1 + 5C_1 \times 5C_1 = (5 \times 5) + (5 \times 5) = 50$$

$$P(E1) = 50 / 100 = 1 / 2$$

$$(iii) n(S) = {}^{10}C_1 \times {}^9C_1 = 10 \times 9 = 90$$

E2 $\rightarrow$ getting an odd

5-even 5-odd

$$n(E2) = 5C_1 \times 5C_1 + 5C_1 \times 5C_1 = 25 + 25 = 50$$

$$P(E2) = 50 / 90 = 5 / 9$$

4) Find i, $P(A \cup B)$ ii, $P(A' \cap B)$ iii, $P(A \cap B')$ iv, $P(A' \cap B')$
if $P(A) = 1/2$, $P(B) = 1/3$, $P(A \cap B) = 1/5$

Answer

$$\begin{aligned} i. P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 1/2 + 1/3 - 1/5 \\ &= (3 + 2)/6 - 1/5 = (5/6) - (1/5) \\ &= (25 - 6)/30 = 19 / 30 \end{aligned}$$

$$\begin{aligned} ii. P(A' \cap B) &= P(B) - P(A \cap B) \\ &= (1/3) - (1/5) \\ &= (5 - 3)/15 = 2 / 15 \end{aligned}$$

$$\begin{aligned} iii. P(A \cap B') &= P(A) - P(A \cap B) \\ &= (1/2) - (1/5) = (5 - 2)/10 = 3 / 10 \end{aligned}$$

$$\begin{aligned} iv. P(A' \cap B') &= P[(A \cup B)'] = 1 - P(A \cup B) \\ &= 1 - \frac{19}{30} = \frac{11}{30} \end{aligned}$$

